

Reference guide: Python concepts from Course 7

Google Cybersecurity Certificate

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Comments

The following syntax is used to create a comment. (A comment is a note programmers make about the intention behind their code.)

#

Starts a line that contains a Python comment

```
# Print approved usernames
```

Contains a comment that indicates the purpose of the code that follows it is to print approved usernames

""" (documentation strings)

Starts and ends a multi-line string that is often used as a Python comment; multi-line comments are used when you need more than 79 characters in a single comment

```
"""
```

```
The estimate_attempts() function takes in a monthly  
login attempt total and a number of months and  
returns their product.
```

```
"""
```

Contains a multi-line comment that indicates the purpose of the `estimate_attempts()` function

Conditional statements

The following keywords and operators are used in conditional statements.

if

Starts a conditional statement

```
if device_id != "1a858zn":
```

Starts a conditional statement that evaluates whether the `device_id` variable contains a value that is not equal to `"1a858zn"`

```
if user in approved_list:
```

Starts a conditional statement that evaluates if the `user` variable contains a value that is also found in the `approved_list` variable

elif

Precedes a condition that is only evaluated when previous conditions evaluate to `False`; previous conditions include the condition in the `if` statement, and when applicable, conditions in other `elif` statements

```
elif status == 500:
```

When previous conditions evaluate to `False`, evaluates if the `status` variable contains a value that is equal to `500`

else

Precedes a code section that only evaluates when all conditions that precede it within the conditional statement evaluate to `False`; this includes the condition in the `if` statement, and when applicable, conditions in `elif` statements

```
else:
```

When previous conditions evaluate to `False`, Python evaluates this `else` statement

and

Requires both conditions on either side of the operator to evaluate to `True`

```
if username == "bmoreno" and login_attempts < 5:
```

Evaluates to `True` if the value in the `username` variable is equal to `"bmoreno"` and the value in the `login_attempts` variable is less than `5`

or

Requires only one of the conditions on either side of the operator to evaluate to `True`

```
if status == 100 or status == 102:
```

Evaluates to `True` if the value in the `status` variable is equal to `100` or the value in the `status` variable is equal to `102`

not

Negates a given condition so that it evaluates to `False` if the condition is `True` and to `True` if it is `False`

```
if not account_status == "removed"
```

Evaluates to `False` if the value in the `account_status` variable is equal to `"removed"` and evaluates to `True` if the value is the `account_status` variable is not equal to `"removed"`

Iterative statements

The following keywords are used in iterative statements.

for

Signals the beginning of a `for` loop; used to iterate through a specified sequence

```
for username in ["bmoreno", "tshah", "elarson"]:
```

Signals the beginning of a `for` loop that iterates through the sequence of elements in the list `["bmoreno", "tshah", "elarson"]` using the loop variable `username`

```
for i in range(10):
```

Signals the beginning of a `for` loop that iterates through a sequence of numbers created by `range(10)` using the loop variable `i`

while

Signals the beginning of a `while` loop; used to iterate based on a condition

```
while login_attempts < 5:
```

Signals the beginning of a `while` loop that will iterate as long as the condition that the value of `login_attempts` is less than `5` evaluates to `True`

break

Used to break out of a loop

continue

Used to skip a loop iteration and continue with the next one

User-defined functions

The following keywords are used when creating user-defined functions.

def

Placed before a function name to define a function

```
def greet_employee():
```

Defines the `greet_employee()` function

```
def calculate_fails(total_attempts, failed_attempts):
```

Defines the `calculate_fails()` function, which includes the two parameters of `total_attempts` and `failed_attempts`

return

Used to return information from a function; when Python encounters this keyword, it exits the function after returning the information

```
def calculate_fails(total_attempts, failed_attempts):  
    fail_percentage = failed_attempts / total_attempts  
    return fail_percentage
```

Returns the value of the `fail_percentage` variable from the `calculate_fails()` function

Built-in functions

The following built-in functions are commonly used in Python.

print()

Outputs a specified object to the screen

```
print("login success")
```

Outputs the string `"login success"` to the screen

```
print(9 < 7)
```

Outputs the Boolean value of `False` to the screen after evaluating whether the integer `9` is less than the integer `7`

type()

Returns the data type of its input

```
print(type(51.1))
```

Returns the data type of float for the input of `51.1`

```
print(type(True))
```

Returns the data type of Boolean for the input of `True`

range()

Generates a sequence of numbers

```
range(0, 5, 1)
```

Generates a sequence with a start point of `0`, a stop point of `5`, and an increment of `1`; because the start point is inclusive but the stop point is exclusive, the generated sequence is `0, 1, 2, 3, and 4`

```
range(5)
```

Generates a sequence with a stop point of `5`; when the start point is not specified, it is set at the default value of `0`, and when the increment is not specified, it is set at the default value of `1`; the generated sequence is `0, 1, 2, 3, and 4`

max()

Returns the largest numeric input passed into it

```
print(max(10, 15, 5))
```

Returns `15` and outputs this value to the screen

min()

Returns the smallest numeric input passed into it

```
print(min(10, 15, 5))
```

Returns 5 and outputs this value to the screen

sorted()

Sorts the components of a list (or other iterable)

```
print(sorted([10, 15, 5]))
```

Sorts the elements of the list from smallest to largest and outputs the sorted list of [5, 10, 15] to the screen

```
print(sorted(["bmoreno", "tshah", "elarson"]))
```

Sorts the elements in the list in alphabetical order and outputs the sorted list of ["bmoreno", "elarson", "tshah"] to the screen

str()

Converts the input object to a string

```
str(10)
```

Converts the integer 10 to the string "10"

len()

Returns the number of elements in an object

```
print(len("security"))
```

Returns and displays 8, the number of characters in the string "security"

Importing modules and libraries

The following keyword is used to import a module from the Python Standard Library or to import an external library that has already been installed.

import

Searches for a module or library in a system and adds it to the local Python environment

```
import statistics
```

Imports the `statistics` module and all of its functions from the Python Standard Library

```
from statistics import mean
```

Imports the `mean()` function of the `statistics` module from the Python Standard Library

```
from statistics import mean, median
```

Imports the `mean()` and `median()` functions of the `statistics` module from the Python Standard Library

String methods

The following methods can be applied to strings in Python.

.upper()

Returns a copy of the string in all uppercase letters

```
print("Security".upper())
```

Returns and displays a copy of the string `"Security"` as `"SECURITY"`

.lower()

Returns a copy of the string in all lowercase letters

```
print("Security".lower())
```

Returns and displays a copy of the string `"Security"` as `"security"`

.index()

Finds the first occurrence of the input in a string and returns its location

```
print("Security".index("c"))
```

Finds the first occurrence of the character "c" in the string "Security" and returns and displays its index of 2

List methods

The following methods can be applied to lists in Python.

.insert()

Adds an element in a specific position inside the list

```
username_list = ["elarson", "fgarcia", "tshah"]  
username_list.insert(2, "wjaffrey")
```

Adds the element "wjaffrey" at index 2 to the username_list; the list becomes ["elarson", "fgarcia", "wjaffrey", "tshah"]

.remove()

Removes the first occurrence of a specific element inside a list

```
username_list = ["elarson", "bmoreno", "wjaffrey", "tshah"]  
username_list.remove("elarson")
```

Removes the element "elarson" from the username_list; the list becomes ["fgarcia", "wjaffrey", "tshah"]

.append()

Adds input to the end of a list

```
username_list = ["bmoreno", "wjaffrey", "tshah"]  
username_list.append("btang")
```

Adds the element "btang" to the end of the username_list; the list becomes ["fgarcia", "wjaffrey", "tshah", "btang"]

.index()

Finds the first occurrence of an element in a list and returns its index

```
username_list = ["bmoreno", "wjaffrey", "tshah", "btang"]  
print(username_list.index("tshah"))
```

Finds the first occurrence of the element "tshah" in the `username_list` and returns and displays its index of 2

Additional syntax for working with strings and lists

The following syntax is useful when working with strings and lists.

+ (concatenation)

Combines two strings or lists together

```
device_id = "IT"+"nwp12"
```

Combines the string "IT" with the string "nwp12" and assigns the combined string of "ITnwp12" to the variable `device_id`

```
users = ["elarson", "bmoreno"] + ["tshah", "btang"]
```

Combines the list ["elarson", "bmoreno"] with the list ["tshah", "btang"] and assigns the combined list of ["elarson", "bmoreno", "tshah", "btang"] to the variable `users`

[] (bracket notation)

Uses indices to extract parts of a string or list

```
print("h32rb17"[0])
```

Extracts the character at index 0, which is ("h"), from the string "h32rb17"

```
print("h32rb17"[0:3])
```

Extracts the slice [0:3], which is ("h32"), from the string "h32rb17"; the first index in the slice (0) is included in the slice but the second index in the slice (3) is excluded

```
username_list = ["elarson", "fgarcia", "tshah"]  
print(username_list[2])
```

Extracts the element at index 2, which is ("tshah"), from the username_list

Regular expressions

The following `re` module function and regular expression symbols are useful when searching for patterns in strings.

`re.findall()`

Returns a list of matches to a regular expression

```
import re  
re.findall("a53", "a53-32c .E")
```

Returns a list of matches to the regular expression pattern "a53" in the string "a53-32c .E"; returns the list ["a53"]

`\w`

Matches with any alphanumeric character; also matches with the underscore (`_`)

```
import re  
re.findall("\w", "a53-32c .E")
```

Returns a list of matches to the regular expression pattern "`\w`" in the string "a53-32c .E"; matches to any alphanumeric character and returns the list ["a", "5", "3", "3", "2", "c", "E"]

`.`

Matches to all characters, including symbols

```
import re  
re.findall(".", "a53-32c .E")
```

Returns a list of matches to the regular expression pattern "." in the string "a53-32c .E"; matches to all characters and returns the list ["a", "5", "3", "-", "3", "2", "c", " ", ".", "E"]

\d

Matches to all single digits

```
import re
re.findall("\d", "a53-32c .E")
```

Returns a list of matches to the regular expression pattern "\d" in the string "a53-32c .E"; matches to all single digits and returns the list ["5", "3", "3", "2"]

\s

Matches to all single spaces

```
import re
re.findall("\s", "a53-32c .E")
```

Returns a list of matches to the regular expression pattern "\s" in the string "a53-32c .E"; matches to all single spaces and returns the list [" "]

\.

Matches to the period character

```
import re
re.findall("\.", "a53-32c .E")
```

Returns a list of matches to the regular expression pattern "\." in the string "a53-32c .E"; matches to all instances of the period character and returns the list ["."]

+

Represents one or more occurrences of a specific character

```
import re
re.findall("\w+", "a53-32c .E")
```

Returns a list of matches to the regular expression pattern "\w+" in the string "a53-32c .E"; matches to one or more occurrences of any alphanumeric character and returns the list ["a53", "32c", "E"]

Represents, zero, one or more occurrences of a specific character

```
import re
re.findall("\w*", "a53-32c .E")
```

Returns a list of matches to the regular expression pattern "`\w*`" in the string "a53-32c .E"; matches to zero, one or more occurrences of any alphanumeric character and returns the list `["a53", " ", "32c", " ", " ", " ", "E"]`

{ }

Represents a specified number of occurrences of a specific character; the number is specified within the curly brackets

```
import re
re.findall("\w{3}", "a53-32c .E")
```

Returns a list of matches to the regular expression pattern "`\w{3}`" in the string "a53-32c .E"; matches to exactly three occurrences of any alphanumeric character and returns the list `["a53", "32c"]`

File operations

The following functions, methods, and keywords are used with operations involving files.

with

Handles errors and manages external resources

```
with open("logs.txt", "r") as file:
```

Used to handle errors and manage external resources while opening a file; the variable `file` stores the file information while inside of the `with` statement; manages resources by closing the file after exiting the `with` statement

open ()

Opens a file in Python

```
with open("login_attempts.txt", "r") as file:
```

Opens the file `"login_attempts.txt"` in order to read it `("r")`
`with open("update_log.txt", "w") as file:`
Opens the file `"update_log.txt"` into the variable `file` in order to write over its contents `("w")`

`with open(import_file, "a") as file:`
Opens the file assigned to the `import_file` variable into the variable `file` in order to append information to the end of it `("a")`

as

Assigns a variable that references another object

`with open("logs.txt", "r") as file:`
Assigns the `file` variable to reference the output of the `open()` function

.read()

Converts files into strings; returns the content of an open file as a string by default

`with open("login_attempts.txt", "r") as file:`
`file_text = file.read()`
Converts the file object referenced in the `file` variable into a string and then stores this string in the `file_text` variable

.write()

Writes string data to a specified file

`with open("access_log.txt", "a") as file:`
`file.write("jrafael")`
Writes the string `"jrafael"` to the `"access_log.txt"` file; because the second argument in the call to the `open()` function is `"a"`, this string is appended to the end of the file

Parsing

The following methods are useful when parsing data.

`.split()`

Converts a string into a list; separates the string based on the character that is passed in as an argument; if an argument is not passed in, it will separate the string each time it encounters whitespace characters such as a space or return

```
approved_users = "elarson,bmoreno,tshah".split(",")
```

Converts the string "elarson,bmoreno,tshah" into the list

["elarson", "bmoreno", "tshah"] by splitting the string into a separate list element at each occurrence of the "," character

```
removed_users = "wjaffrey jsoto abernard".split()
```

Converts the string "wjaffrey jsoto abernard" into the list

["wjaffrey", "jsoto", "abernard"] by splitting the string into a separate list element at each space

`.join()`

Concatenates the elements of an iterable into a string; takes the iterable to be concatenated as an argument; is appended to a character that will separate each element once they are joined into a string

```
approved_users = ",".join(["elarson", "bmoreno", "tshah"])
```

Concatenates the elements of the list ["elarson", "bmoreno", "tshah"] into the string "elarson,bmoreno,tshah", separating each element with the "," character within the string